# Solution: Problem Challenge 3: Count of Structurally Unique Binary Search Trees

## Problem Statement

Given a number 'n', write a function to return the count of structurally unique Binary Search Trees (BST) that can store values 1 to 'n'.

**Example 1:**

```
Input: 2
Output: 2
Explanation: As we saw in the previous problem, there are 2 unique BSTs
storing numbers from 1-2.
```

**Example 2:**

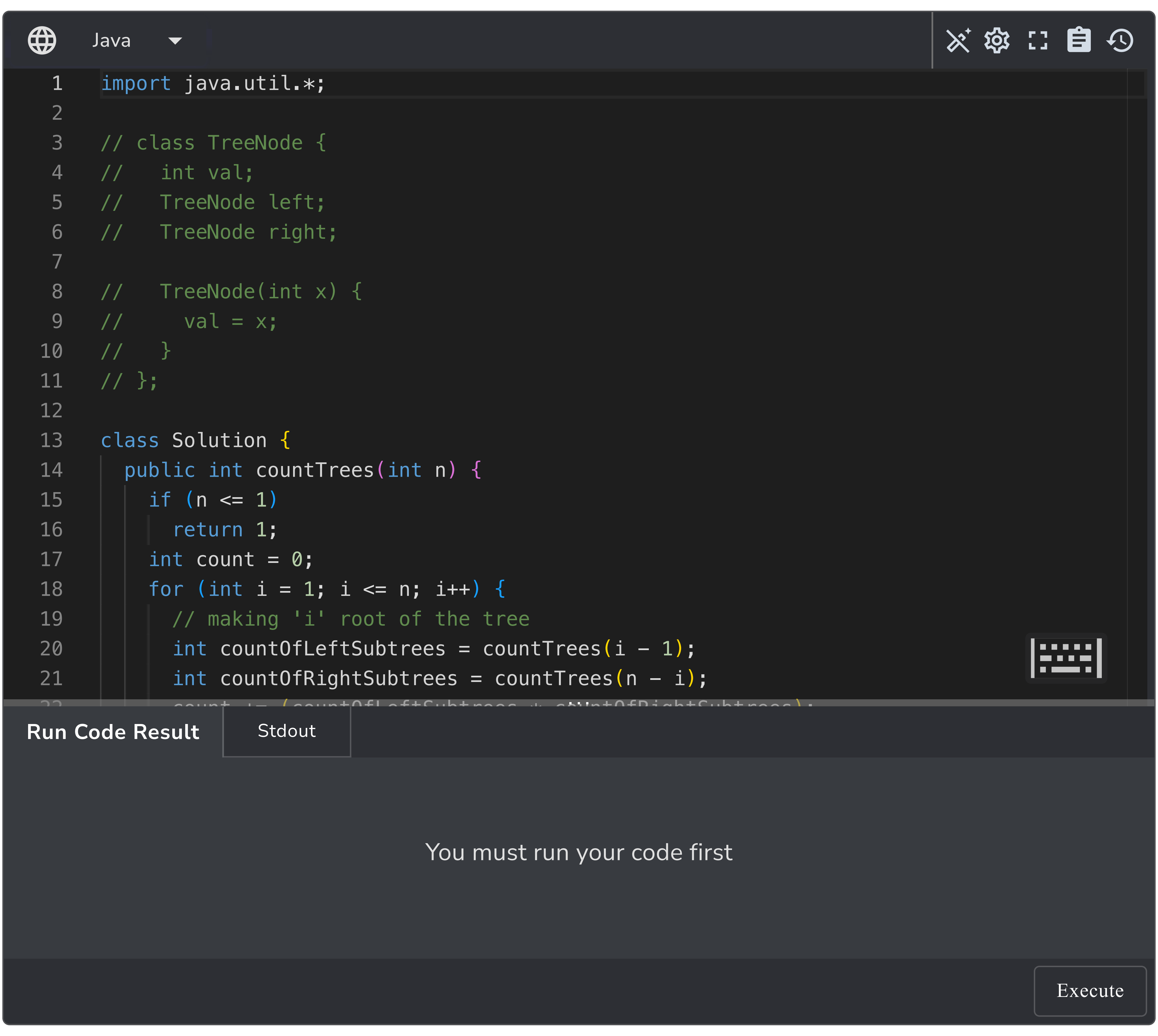
```
Input: 3
Output: 5
Explanation: There will be 5 unique BSTs that can store numbers from 1
to 3.
```

## Solution

This problem is similar to `Structurally Unique Binary Search Trees`. Following a similar approach, we can iterate from 1 to 'n' and consider each number as the root of a tree and make two recursive calls to count the number of left and right sub-trees.

## Code

Here is what our algorithm will look like:



```java
import java.util.*;

// class TreeNode {
//   int val;
//   TreeNode left;
//   TreeNode right;

//   TreeNode(int x) {
//     val = x;
//   }
// };

class Solution {
  public int countTrees(int n) {
    if (n <= 1)
      return 1;
    int count = 0;
    for (int i = 1; i <= n; i++) {
      // making 'i' root of the tree
      int countOfLeftSubtrees = countTrees(i - 1);
      int countOfRightSubtrees = countTrees(n - i);
```

You must run your code first

### Time complexity

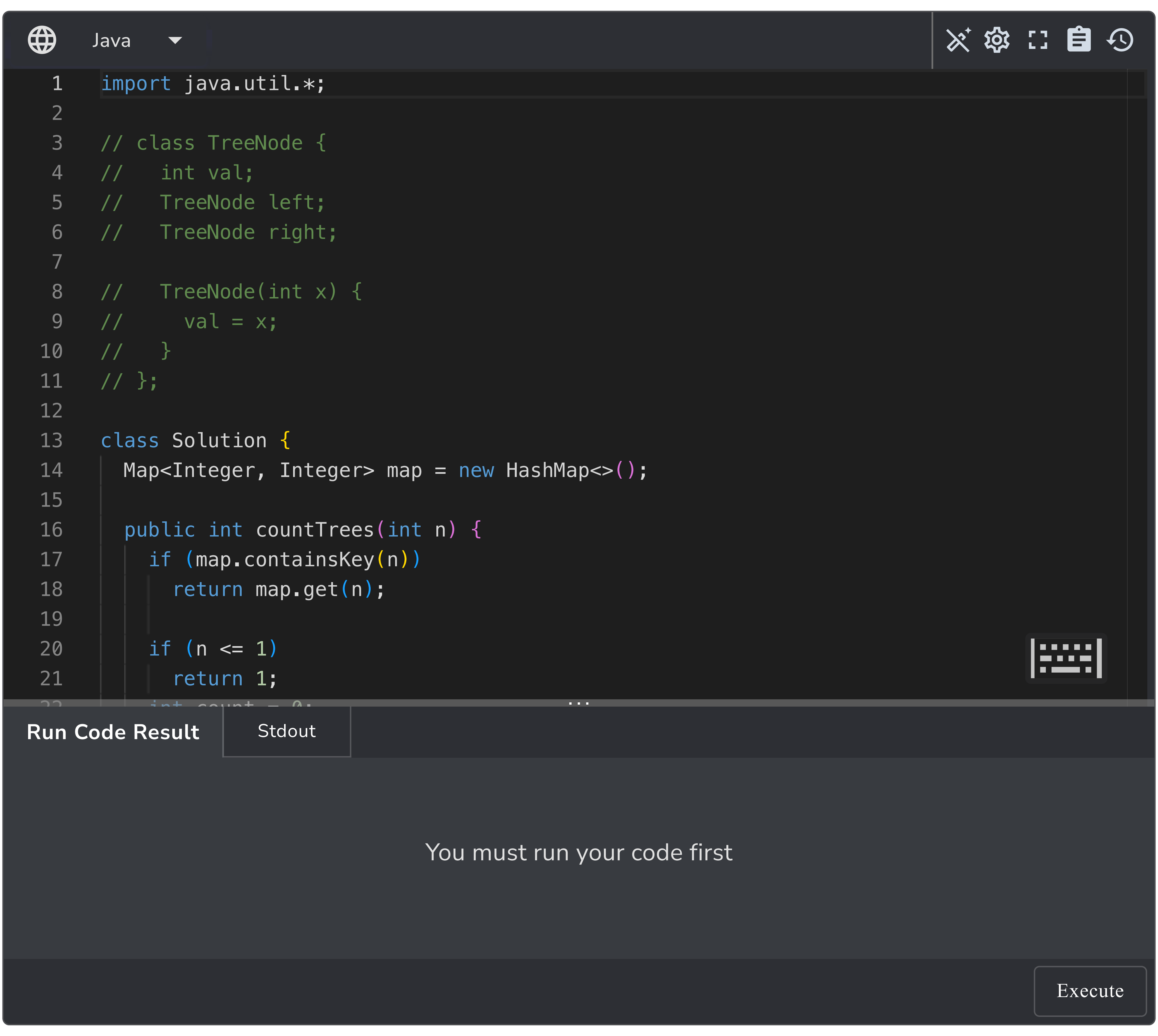
The time complexity of this algorithm will be exponential and will be similar to Balanced Parentheses. Estimated time complexity will be $O(n * 2^n)$ but the actual time complexity $O(4^n/\sqrt{n})$ is bounded by the Catalan number and is beyond the scope of a coding interview. See more details here.

### Space complexity

The space complexity of this algorithm will be exponential too, estimated $O(2^n)$ but the actual will be $O(4^n/\sqrt{n})$.

## Memoized version

Our algorithm has overlapping subproblems as our recursive call will be evaluating the same sub-expression multiple times. To resolve this, we can use memoization and store the intermediate results in a HashMap. In each function call, we can check our map to see if we have already evaluated this sub-expression before. Here is the memoized version of our algorithm:



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// class TreeNode {
//   int val;
//   TreeNode left;
//   TreeNode right;

//   TreeNode(int x) {
//     val = x;
//   }
// };

class Solution {
  Map<Integer, Integer> map = new HashMap<>();

  public int countTrees(int n) {
    if (map.containsKey(n))
      return map.get(n);

    if (n <= 1)
      return 1;
```

You must run your code first

The time complexity of the memoized algorithm will be $O(n^2)$, since we are iterating from '1' to 'n' and ensuring that each sub-problem is evaluated only once. The space complexity will be $O(n)$ for the memoization map.